THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $3\frac{1}{2}$	Convert both to improper fractions first.
2	**(c)** 2	Improper form, then multiply by the reciprocal.
3	**(c)** 1	Work from left to right.
4	**(b)** 21	Number = part ÷ fraction.
5	**(b)** $2\frac{1}{2}$	Time = $1 \div$ rate.
6	**(b)** $\frac{3}{4} \div \frac{1}{8}$	Compare by computing.
7	**(a)** 105	Distance = rate × quantity.
8	**(c)** 3	Convert both to improper fractions first.
9	**(a)** 6	Improper form, then multiply by the reciprocal.
10	**(d)** $\frac{2}{3}$	Work from left to right.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Left to right: $\frac{2}{3} \times \frac{9}{10} = \frac{3}{5}$; Then divide: $\frac{3}{5} \times \frac{5}{3}$ (1); Answer 1 (1).	1+1
12	Number = $27 \div \frac{3}{4}$; = 36 (1); $\frac{1}{4}$ of it = 9 (1).	1+1
13	Product = $1\frac{1}{2}$; Compare with 2\frac{1}{2} (1); The product is smaller; Because the multiplier is less than 1 (1).	1+1
14	Left to right: $\frac{5}{6} \times \frac{4}{15} = \frac{2}{9}$; Then divide: $\frac{2}{9} \times \frac{9}{2}$ (1); Answer 1 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = $\frac{3}{2} \times \frac{9}{4}$; = $3\frac{3}{8}$ m^2 (1); Perimeter = $2(1\frac{1}{2} + 2\frac{1}{4})$; = $2 \times 3\frac{3}{4}$ (1); = $7\frac{1}{2}$ m (1).	1+1+1
16	Spent = $\frac{1}{4} \times 240 = 60$; Left = $240 - 60 = 180$ (1); Each = $180 \div 6$ (1); = ₹30 (1).	1+1+1
17	Bracket: $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ (1); Multiply: $\frac{5}{6} \times \frac{6}{5}$ (1); = 1 (1).	1+1+1
	OR: Area = $\frac{7}{3} \times \frac{3}{2}$; = $3\frac{1}{2}$ m^2 (1); Perimeter = $2(2\frac{1}{3} + 1\frac{1}{2})$; = $2 \times 3\frac{5}{6}$ (1); = $7\frac{2}{3}$ m (1).	1+1+1
18	Spent = $\frac{2}{5} \times 360 = 144$; Left = $360 - 144 = 216$ (1); Each = $216 \div 9$ (1); = ₹24 (1).	1+1+1
	OR: Bracket: $\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$ (1); Multiply: $\frac{11}{12} \times \frac{12}{7}$ (1); = $1\frac{4}{7}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day 1 = $\frac{1}{4} \times 240 = 60$ m^2 (1); Remainder = 180 m^2 (1); Day 2 = $\frac{2}{3} \times 180 = 120$ m^2 (1); Day 3 = $180 - 120$ (1); = 60 m^2 (1).	1+1+1+1+1
	OR: Syrup = $\frac{2}{5} \times \frac{5}{2} = 1$ L (1); Water = $2\frac{1}{2} - 1 = 1\frac{1}{2}$ L (1); Glasses = $2\frac{1}{2} \div \frac{1}{4}$ (1); = 10 (1); Full glasses: 10 (1).	1+1+1+1+1
20	Day 1 = $\frac{2}{5} \times 180 = 72$ m^2 (1); Remainder = 108 m^2 (1); Day 2 = $\frac{1}{3} \times 108 = 36$ m^2 (1); Day 3 = $108 - 36$ (1); = 72 m^2 (1).	1+1+1+1+1
	OR: Syrup = $\frac{3}{7} \times \frac{7}{4} = \frac{3}{4}$ L (1); Water = $1\frac{3}{4} - \frac{3}{4} = 1$ L (1); Glasses = $1\frac{3}{4} \div \frac{1}{5}$ (1); = $8\frac{3}{4}$ (1); Full glasses: 8 (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change ÷ to ×.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a + sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30	Number = part ÷ fraction.
2	**(d)** $2\frac{2}{3}$	Time = 1÷ rate.
3	**(b)** 1	Compare by computing.
4	**(b)** 94	Distance = rate × quantity.
5	**(b)** $5\frac{1}{3}$	Convert both to improper fractions first.
6	**(d)** 4	Improper form, then multiply by the reciprocal.
7	**(c)** $\frac{3}{4}$	Work from left to right.
8	**(b)** 40	Number = part ÷ fraction.
9	**(a)** $2\frac{1}{4}$	Time = 1÷ rate.
10	**(b)** 1	Compare by computing.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number = $18 \div \frac{2}{5}$; = 45 (1); $\frac{3}{5}$ of it = 27 (1).	1+1
12	Product = $1\frac{1}{2}$; Compare with 1\frac{1}{3} (1); The product is greater; Because the multiplier is more than 1 (1).	1+1
13	Left to right: $\frac{7}{8} \times \frac{16}{21} = \frac{2}{3}$; Then divide: $\frac{2}{3} \times \frac{3}{4}$ (1); Answer $\frac{1}{2}$ (1).	1+1
14	Number = $35 \div \frac{5}{6}$; = 42 (1); $\frac{1}{6}$ of it = 7 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = $\frac{7}{2} \times \frac{7}{3}$; = $8\frac{1}{6}$ m^2 (1); Perimeter = $2(3\frac{1}{2} + 2\frac{1}{3})$; = $2 \times 5\frac{5}{6}$ (1); = $11\frac{2}{3}$ m (1).	1+1+1
16	Spent = $\frac{1}{3} \times 450 = 150$; Left = $450 - 150 = 300$ (1); Each = $300 \div 5$ (1); = ₹60 (1).	1+1+1
17	Bracket: $\frac{2}{3} + \frac{3}{5} = 1\frac{4}{15}$ (1); Multiply: $1\frac{4}{15} \times \frac{15}{19}$ (1); = 1 (1).	1+1+1
	OR: Area = $\frac{7}{4} \times \frac{5}{2}$; = $4\frac{3}{8}$ m^2 (1); Perimeter = $2(1\frac{3}{4} + 2\frac{1}{2})$; = $2 \times 4\frac{1}{4}$ (1); = $8\frac{1}{2}$ m (1).	1+1+1
18	Spent = $\frac{3}{8} \times 560 = 210$; Left = $560 - 210 = 350$ (1); Each = $350 \div 7$ (1); = ₹50 (1).	1+1+1
	OR: Bracket: $\frac{5}{8} + \frac{1}{4} = \frac{7}{8}$ (1); Multiply: $\frac{7}{8} \times \frac{16}{21}$ (1); = $\frac{2}{3}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day 1 = $\frac{3}{10} \times 300 = 90$ m^2 (1); Remainder = 210 m^2 (1); Day 2 = $\frac{5}{7} \times 210 = 150$ m^2 (1); Day 3 = $210 - 150$ (1); = 60 m^2 (1).	1+1+1+1+1
	OR: Syrup = $\frac{3}{10} \times \frac{10}{3} = 1$ L (1); Water = $3\frac{1}{3} - 1 = 2\frac{1}{3}$ L (1); Glasses = $3\frac{1}{3} \div \frac{1}{6}$ (1); = 20 (1); Full glasses: 20 (1).	1+1+1+1+1
20	Day 1 = $\frac{1}{6} \times 210 = 35$ m^2 (1); Remainder = 175 m^2 (1); Day 2 = $\frac{3}{5} \times 175 = 105$ m^2 (1); Day 3 = $175 - 105$ (1); = 70 m^2 (1).	1+1+1+1+1
	OR: Syrup = $\frac{3}{8} \times \frac{8}{3} = 1$ L (1); Water = $2\frac{2}{3} - 1 = 1\frac{2}{3}$ L (1); Glasses = $2\frac{2}{3} \div \frac{1}{4}$ (1); = $10\frac{2}{3}$ (1); Full glasses: 10 (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
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- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change ÷ to ×.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a + sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 85	Distance = rate × quantity.
2	**(b)** 6	Convert both to improper fractions first.
3	**(d)** 2	Improper form, then multiply by the reciprocal.
4	**(d)** 1	Work from left to right.
5	**(b)** 28	Number = part ÷ fraction.
6	**(c)** $1\frac{2}{5}$	Time = $1 \div$ rate.
7	**(d)** $\frac{7}{3}$	Compare by computing.
8	**(b)** 58	Distance = rate × quantity.
9	**(c)** $\frac{2}{5}$	Work from left to right.
10	**(d)** $3\frac{1}{3}$	Time = $1 \div$ rate.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Product $= 2\frac{1}{6}$; Compare with 3\frac{1}{4} (1); The product is smaller; Because the multiplier is less than 1 (1).	1+1
12	Left to right: $\frac{3}{4} \times \frac{8}{9} = \frac{2}{3}$; Then divide: $\frac{2}{3} \times \frac{6}{5}$ (1); Answer $\frac{4}{5}$ (1).	1+1
13	Number $= 32 \div \frac{4}{9}$; $= 72$ (1); $\frac{5}{9}$ of it $= 40$ (1).	1+1
14	Product $= 3$; Compare with 2\frac{2}{5} (1); The product is greater; Because the multiplier is more than 1 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{8}{3} \times \frac{5}{4}$; $= 3\frac{1}{3}$ m^2 (1); Perimeter $= 2(2\frac{2}{3} + 1\frac{1}{4})$; $= 2 \times 3\frac{11}{12}$ (1); $= 7\frac{5}{6}$ m (1).	1+1+1
16	Spent $= \frac{2}{7} \times 420 = 120$; Left $= 420 - 120 = 300$ (1); Each $= 300 \div 10$ (1); $=$ ₹30 (1).	1+1+1
17	Bracket: $\frac{1}{4} + \frac{2}{5} = \frac{13}{20}$ (1); Multiply: $\frac{13}{20} \times \frac{20}{9}$ (1); $= 1\frac{4}{9}$ (1).	1+1+1
	OR: Area $= \frac{9}{2} \times \frac{10}{3}$; $= 15$ m^2 (1); Perimeter $= 2(4\frac{1}{2} + 3\frac{1}{3})$; $= 2 \times 7\frac{5}{6}$ (1); $= 15\frac{2}{3}$ m (1).	1+1+1
18	Spent $= \frac{4}{9} \times 630 = 280$; Left $= 630 - 280 = 350$ (1); Each $= 350 \div 7$ (1); $=$ ₹50 (1).	1+1+1
	OR: Bracket: $\frac{3}{5} + \frac{1}{10} = \frac{7}{10}$ (1); Multiply: $\frac{7}{10} \times \frac{10}{7}$ (1); $= 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day 1 $= \frac{3}{8} \times 320 = 120$ m^2 (1); Remainder $= 200$ m^2 (1); Day 2 $= \frac{4}{5} \times 200 = 160$ m^2 (1); Day 3 $= 200 - 160$ (1); $= 40$ m^2 (1).	1+1+1+1+1
	OR: Syrup $= \frac{4}{9} \times \frac{3}{2} = \frac{2}{3}$ L (1); Water $= 1\frac{1}{2} - \frac{2}{3} = \frac{5}{6}$ L (1); Glasses $= 1\frac{1}{2} \div \frac{1}{3}$ (1); $= 4\frac{1}{2}$ (1); Full glasses: 4 (1).	1+1+1+1+1
20	Day 1 $= \frac{2}{9} \times 270 = 60$ m^2 (1); Remainder $= 210$ m^2 (1); Day 2 $= \frac{3}{7} \times 210 = 90$ m^2 (1); Day 3 $= 210 - 90$ (1); $= 120$ m^2 (1).	1+1+1+1+1
	OR: Syrup $= \frac{5}{11} \times \frac{11}{4} = 1\frac{1}{4}$ L (1); Water $= 2\frac{3}{4} - 1\frac{1}{4} = 1\frac{1}{2}$ L (1); Glasses $= 2\frac{3}{4} \div \frac{1}{8}$ (1); $= 22$ (1); Full glasses: 22 (1).	1+1+1+1+1

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